

Phase-Wise AI Prompt Documentation

Project Title

ExpirySense – SSL Certificate Expiry Watcher with AI Agent

Phase 1 – Project Ideation and Requirement Analysis

Objective

Identify a cybersecurity problem and define project requirements.

AI Prompt Used

"Suggest innovative cybersecurity projects that combine SSL certificate monitoring, AI-powered recommendations, and full-stack development for technical interview demonstrations."

AI Contribution

- Proposed SSL certificate monitoring as the core problem.
- Suggested AI-powered remediation recommendations.
- Identified dashboard and reporting requirements.
- Recommended integration of AI agents and MCP architecture.

Outcome

Project idea finalized as:

ExpirySense – SSL Certificate Expiry Watcher with AI Agent.

Phase 2 – System Architecture Design

Objective

Design overall architecture of the application.

AI Prompt Used

"Design a scalable architecture for an SSL certificate monitoring system using FastAPI, SQLite, Groq API, and MCP integration."

AI Contribution

- Designed frontend-backend architecture.
- Suggested FastAPI framework.
- Recommended SQLite database.

- Proposed MCP-based communication.
- Defined AI Agent layer.

Outcome

System architecture finalized.

Phase 3 – Database Design

Objective

Design database schema.

AI Prompt Used

"Generate an SQLite schema for storing SSL certificate scan results including hostname, issuer, expiry date, risk level, and scan history."

AI Contribution

- Generated table structure.
- Recommended fields.
- Suggested storage of scan timestamps.

Outcome

ssl_certificates database schema created.

Phase 4 – SSL Scanning Module Development

Objective

Develop SSL certificate scanning engine.

AI Prompt Used

"Generate Python code using socket and ssl libraries to retrieve SSL certificate details including issuer, expiry date, and TLS version."

AI Contribution

- Suggested SSL scanning workflow.
- Provided certificate extraction logic.
- Recommended error handling strategies.

Outcome

Real-time SSL certificate scanner implemented.

Phase 5 – Frontend Dashboard Development

Objective

Develop user interface and dashboard.

AI Prompt Used

"Create a modern cybersecurity dashboard with certificate statistics, risk indicators, tables, and charts."

AI Contribution

- Suggested dashboard layout.
- Recommended Chart.js visualizations.
- Designed user-friendly workflow.

Outcome

Interactive dashboard created.

Phase 6 – AI Agent Development

Objective

Integrate Artificial Intelligence.

AI Prompt Used

"Design an AI agent using Groq API and Llama 3 that can generate SSL remediation recommendations and business impact analysis."

AI Contribution

- Defined AI Agent workflow.
- Suggested prompt engineering techniques.
- Generated recommendation templates.
- Created report generation logic.

Outcome

AI Agent integrated successfully.

Phase 7 – MCP Integration

Objective

Implement Model Context Protocol.

AI Prompt Used

"Create MCP tools for SSL scanning, certificate analysis, recommendation generation, and report creation."

AI Contribution

- Designed MCP server architecture.
- Defined MCP tools.
- Suggested tool communication workflow.

Outcome

MCP integration completed.

Phase 8 – Report Generation

Objective

Generate downloadable reports.

AI Prompt Used

"Create CSV export functionality and Markdown-based renewal workbook generation for expiring SSL certificates."

AI Contribution

- Generated report formats.
- Created renewal email templates.
- Suggested downloadable report structure.

Outcome

CSV and Markdown export features implemented.

Phase 9 – Testing and Validation

Objective

Verify system functionality.

AI Prompt Used

"Generate test cases for SSL certificate monitoring applications including valid domains, invalid domains, expired certificates, and network failures."

AI Contribution

- Generated functional test cases.
- Suggested validation scenarios.
- Identified edge cases.

Outcome

Application tested successfully.

Phase 10 – Documentation and Deployment

Objective

Prepare documentation and deployment workflow.

AI Prompt Used

"Generate deployment instructions for a FastAPI application integrated with Groq API and SQLite database."

AI Contribution

- Generated deployment steps.
- Suggested hosting platforms.
- Created project documentation outline.

Outcome

Project prepared for deployment and presentation.

AI Tools Used

- Google AI Studio
- Groq API
- Llama3-8B-8192
- MCP Architecture
- Generative AI Prompt Engineering